

College of Industrial Technology
King Mongkut's University of Technology North Bangkok

Seat No.

Final Examination of Semester 2

Year: 2024

Subject: 031021122 Electric Circuits II

Section: 5-6

Date: 10 March 2025

Time: 8.00-10.00

Name: _____ ID: _____ Class: _____

Directions: The test is designed to measure your comprehension. The test is divided into 3 sections. There will be 11 pages (including this page) and they are worth 85 points.

- This exam paper contains no errors. If a suspected error is found, it is the student's discretion to correct it.
- Answer the questions on this test papers.
- Books, documents, and lecture notes are not allowed.
- You must be in the room for one hour after the exam is started and, while taking the exam, you cannot go out except in an emergency case.
- Before leaving, make sure you do not bring this test outside.
- Do not use any electronic communication device.
- Calculators can be used in this test.

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Now begin the test.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

This exam paper is part of a set used to assess student abilities as follows:

		Part 1	Part 2	Part 3
CLO 1	Explain Capacitor and inductor in DC circuit, alternating current (AC) circuits, Sinusoidal alternating waveforms, sinusoidal voltage and current, phase relations, average value, effective (rms) value, basic R, L, and C elements in AC circuits, average power. The power triangle, power-factor correction, three phase generator, Y-connected generator, phase sequence, and star-delta system. Rectangular form, polar form, series and parallel AC circuits, series-parallel AC networks	✓		
CLO 2	Identify the types of capacitor, inductor and resistor in AC networks.	✓	✓	
CLO 3	Analyze the series capacitor, inductor and resistor in AC networks, parallel capacitor, inductor and resistor in AC networks and series-parallel capacitor, inductor and resistor in AC networks using methods of analysis for AC networks		✓	✓
CLO 4	Design the series capacitor, inductor and resistor in AC networks, parallel capacitor, inductor and resistor in AC networks and series-parallel capacitor, inductor and resistor in AC networks using methods of analysis for AC networks		✓	✓

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Part 1 (20 points) True/False

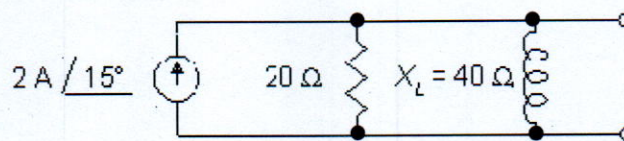
Instruction: Read the following statements, mark "X" for the correct statement, "True" or "False" in the provided answer sheet (see LAST PAGE).

1. Any 2-terminal linear AC network can be replaced by an equivalent circuit of Thevenin's theorem consisting of a voltage source and impedance in series.
2. The current source from Norton equivalent circuit can be obtained from open circuit current.
3. To remove a current source, the current source can be shorted circuit in the superposition theorem.
4. A dependent current source may be controlled by either a current or a voltage from other elements.
5. Apparent power at a resistor is equal to zero.
6. Capacitive element has negative reactive power.
7. The unit of reactive power is in Watt.
8. Total apparent power is the sum of average power and reactive power of each element in series or parallel circuit.
9. The maximum current can be occurred in a resonant series-RLC circuit.
10. The current drawn from the source can be reduced if the total reactive power is reduced.
11. The aim of power factor correction is to make the power factor to be infinity.
12. A purely inductive circuit has a power factor of 0 lagging, meaning all power is reactive.
13. The RMS value of a sinusoidal waveform is always greater than its average value.
14. In a balanced three-phase system, the sum of the three line currents is always zero.
15. The bandwidth of a series RLC circuit decreases when the resistance increases.
16. The phase angle of an RC circuit is always positive.
17. Admittance is the reciprocal of impedance and is measured in Siemens (S).
18. In a parallel resonant circuit, the impedance is minimum at resonance.
19. The real power in an AC circuit is given by $P=VI$ (VA).
20. A leading power factor means the load is capacitive.

Part 2 (40 points) Multiple choice

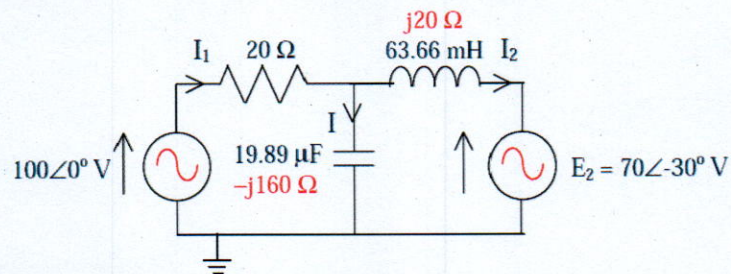
Instruction: Mark the correct answer for the following questions in the provided answer sheet (see LAST PAGE).

- Calculate the apparent power (S) in an AC circuit with a real power (P) of 100 W and a power factor (PF) of 0.8.
 - 100 VA
 - 125 VA
 - 150 VA
 - 175 VA
- If the reactive power (Q) in an AC circuit is 50 VAR and the voltage is 120 V, calculate the power factor.
 - 0.4
 - 0.5
 - 0.6
 - 0.7
- Convert the current source to a voltage source. What is the angle of the current source?



- (a) 41.6° (b) 25.5° (c) 27.1° (d) 24.3°

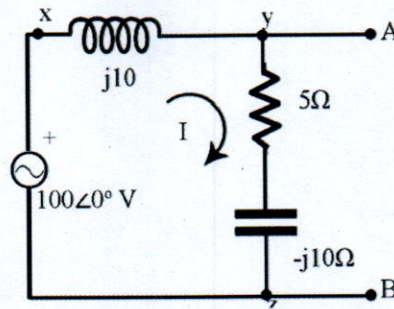
4. From figure below, what is KVL equation at loop E_2 ?



- (a) $7\angle -30^\circ = +j16I_2 - j14I_1$ (b) $5\angle 0^\circ = (1 - j8)I_1 + j8I_2$
 (c) $7\angle -30^\circ = -j16I_1 + j14I_2$ (d) None of the above

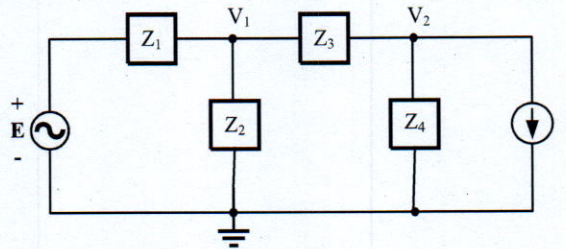
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5. From figure below, what is the Norton equivalent current I_N at A-B terminals?



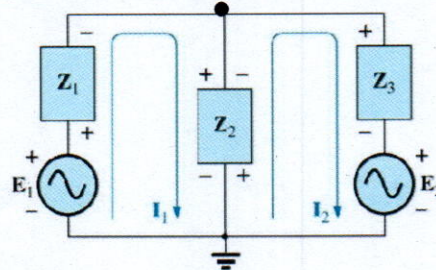
- (a) $20\angle 90^\circ$ A (b) $12\angle 90^\circ$ A
(c) $10\angle -90^\circ$ A (d) $5\angle 90^\circ$ A

6. From the circuit below, what is KCL equation at node V_2 ?



- (a) $(V_1 + E)/Z_1 + V_1/Z_2 + (V_1 - V_2)/Z_3 = 0$
 (b) $(V_2 + V_1)/Z_3 + V_2/Z_4 - I = 0$
 (c) $(V_1 + E)/Z_1 + V_1/Z_2 + (V_1 + V_2)/Z_3 = 0$
 (d) $(V_2 - V_1)/Z_3 + V_2/Z_4 + I = 0$

7. From the circuit below, what is KVL equation at mesh I_1 ?



- (a) $E_1 + Z_1 I_1 + Z_2 (I_1 - I_2) = 0$ (b) $E_1 + Z_1 I_1 + Z_2 (I_2 - I_1) = 0$
(c) $-E_1 + Z_1 I_1 + Z_2 (I_1 - I_2) = 0$ (d) $-E_1 + Z_1 I_1 + Z_2 (I_2 - I_1) = 0$

8. An RL circuit is connected to supply with 230 V and 50 Hz provides $S=500$ VA. The resistor is $20\ \Omega$, what is the inductive reactance (X_L) value?

- (a) $103.89\ \Omega$ (b) $330.7\ \Omega$ (c) $105.8\ \Omega$ (d) $100.5\ \Omega$

9. A motor draws 1 A when supplied from a 110 V AC supply. The power factor is 0.8 lagging. What is the real power consumed by the motor?

- (a) 440 W (b) 550 W (c) 54 W (d) 88 W

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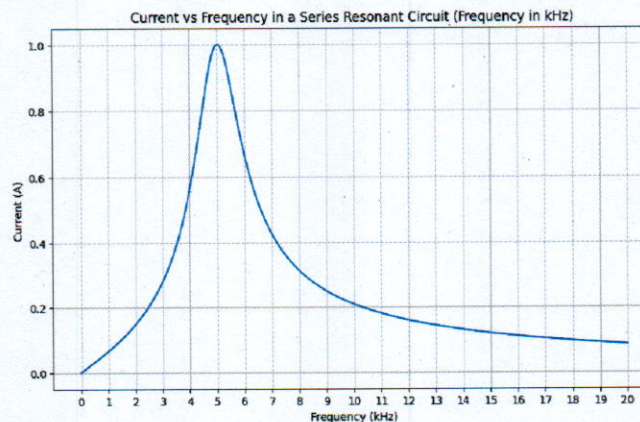
10. Convert the current source to a voltage source. Given $I_s = 2\sin(\omega t + 40^\circ)$ A and parallel impedance $Z = 100 + j0$. What is the new voltage source?
- (a) $V_s = 20 \angle -30^\circ$ (b) $V_s = 200 \angle 40^\circ$ (c) $V_s = 2 \angle 30^\circ$ (d) $V_s = 20 \angle -40^\circ$
11. If $S = 10 + j10$, which of the following is true?
- (a) Power factor is greater than one.
 (b) Inductive reactance must be greater than capacitive reactance.
 (c) It is pure resistive circuit
 (d) Its unit is in VAR
12. If the transformation ratio is smaller than 1, what is the transformer called?
- (a) step down (b) secondary
 (c) primary (d) step up
13. If the transformation ratio is 2.4, and the primary voltage is 12 VDC, what is the secondary voltage?
- (a) 28.8 V (b) 5 V (c) 0 V (d) 0.2 V
14. The load on a 220 V, 60 Hz supply is 10 kW (resistive), 20 kVAR (inductive), and 5 kVAR (capacitive). Find the total apparent power.
- (a) 24 kVA (b) 18.03 kVA
 (c) 21.5 kVA (d) 14 kVA
15. What is the power factor of the problem 14.
- (a) 1.8 (b) 1
 (c) 0.83 (d) 0.55
16. In Problem 14, determine the capacitance needed to bring the circuit to unity power factor.
- (a) 102 μF (b) 821 μF
 (c) 84.6 μF (d) 1004 mF
17. The phase voltage of a Y-connected three-phase generator is 120 V. What is the line voltage?
- (a) 69 V (b) 170 V (c) 208 V (d) 85 V
18. At 60 Hz, a series LCR circuit contains reactance $X_L = 20 \Omega$ and $X_C = 14 \Omega$ and resistance = 10 Ω . What is the impedance at resonant frequency?
- (a) 26.4 Ω (b) 24.4 Ω (c) 10 Ω (d) 20 Ω

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19. What is the resonant frequency of a series circuit consisting of a 100 pF capacitor, a 10 k Ω resistor, and a 1 mH inductor?

- (a) 503 kHz (b) 159 Hz
(c) 3.16 MHz (d) 1.58 MHz

20. From the curve below, describes the characteristics of a series resonant circuit. Determine the resonant frequency and an estimated bandwidth of circuit.



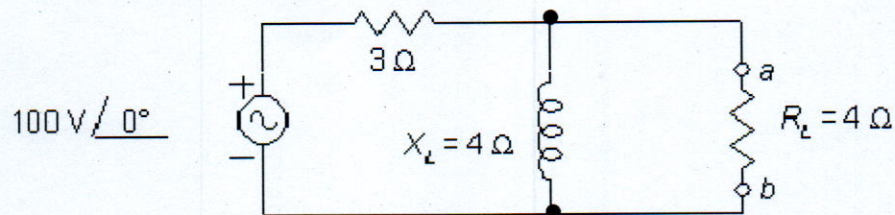
- (a) 5 kHz and 1.6 kHz (b) 5 Hz and 1.6 Hz
(c) 500 Hz and 160 Hz (d) 5 kHz and 2 kHz

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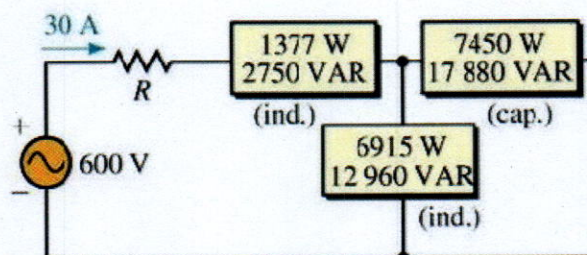
Part 3 (25 points)

Instruction: Show the mathematical expression and answers of following problems.

1. Find the Thévenin equivalent circuit external to the load resistor R_L . What is the power dissipated by the load resistor?

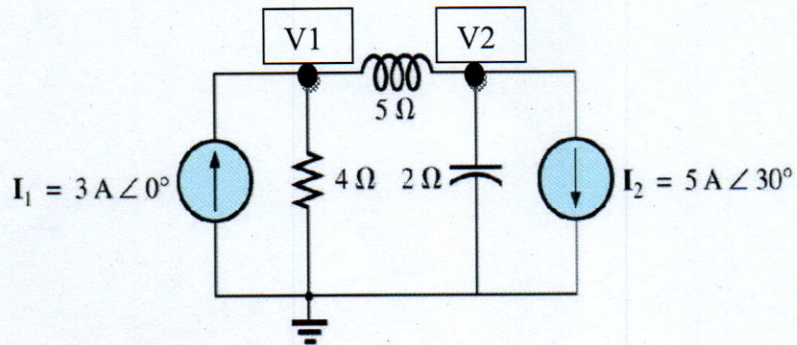


2. Given $R=2.36\ \Omega$. Determine the value of P_T , Q_T and S and draw the power triangle.



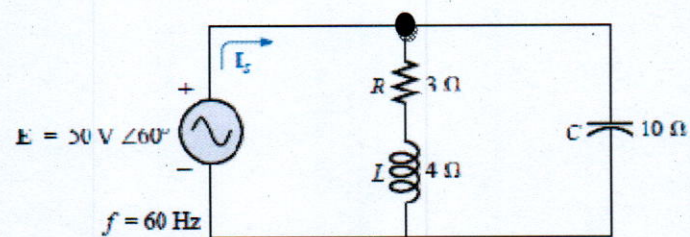
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3. Determine the nodal voltages V_1 and V_2 for the network below.



4. From the network in Figure below;

- (a) Find the total real power P_T (watts), reactive power (Q_T), and apparent power (S),
 (b) Sketch the power triangle.



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5. A balanced Δ load having a $200\ \Omega$ resistance in each leg is connected to a three-phase, three-wire, Δ -connected generator having a line voltage of 220 V. Calculate the magnitude of the following
- (a) phase voltage of the generator.
 - (b) phase voltage of the load.
 - (c) phase current of the load.
 - (d) line current.

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Instructors: Prof. Dr. Titipong Lertwiriaprapa
and Asst. Prof. Dr. Kittisak Phaebua

Answer Sheet

Name: _____ ID: _____

Subject: 031021122 Electric Circuits II

Part 1

	True	False
1.		
2.		
3.		
4.		
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Part 2

	a	b	c	d
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